Amazon EBS — Important Interview Questions & Answers

1.What is Amazon EBS?
Amazon EBS is a block storage service designed to provide persistent storage for EC2 instances.
It can be used to store operating systems, applications, databases, and other data.
Simple ga gurthupettuko:
EBS ante EC2 ki persistent block storage provide chese AWS service.
Eng: EBS is a persistent block storage service used with EC2.

2.What is an EBS Volume?
An EBS Volume is a virtual disk that can be attached to an EC2 instance.
It provides persistent storage for the operating system, applications, and data.
Simple ga gurthupettuko:
EBS Volume ante EC2 ki attach chese virtual hard disk.
Eng: An EBS Volume is a virtual disk that provides persistent storage for EC2.

3.What are EBS Volume Types?
EBS provides different volume types designed for different performance and workload requirements.
Common types include General Purpose SSD, Provisioned IOPS SSD, Throughput Optimized HDD, and Cold HDD.
Simple ga gurthupettuko:
EBS Volume Types ante different performance and workload requirements kosam available unna storage options.
Eng: EBS Volume Types provide different storage performance options for different workloads.

4.What is General Purpose SSD?
General Purpose SSD volumes provide a balance of price and performance for many workloads.
They are commonly used for applications, boot volumes, and general-purpose databases.
Simple ga gurthupettuko:
General Purpose SSD ante normal applications and databases kosam balanced performance iche EBS volume.
Eng: General Purpose SSD provides balanced price and performance for general workloads.

5.What is Provisioned IOPS SSD?
Provisioned IOPS SSD volumes are designed for workloads that require high and consistent I/O performance.
They are commonly used for performance-sensitive databases and applications.
Simple ga gurthupettuko:
Provisioned IOPS SSD ante high and consistent I/O performance kavali ante use chestham.
Eng: Provisioned IOPS SSD is used for workloads requiring high and consistent I/O performance.

6.What is Throughput Optimized HDD?
Throughput Optimized HDD is designed for frequently accessed, throughput-intensive workloads.
It is suitable for workloads where high throughput is more important than very high IOPS.
Simple ga gurthupettuko:
Throughput Optimized HDD ante high data throughput kavalsina workloads kosam use chestham.
Eng: Throughput Optimized HDD is designed for throughput-intensive workloads.

7.What is Cold HDD?
Cold HDD is a lower-cost HDD volume designed for less frequently accessed workloads.
It is useful when storage cost is more important than high performance.
Simple ga gurthupettuko:
Cold HDD ante infrequently accessed data ni low cost lo store cheyyadaniki use chestham.
Eng: Cold HDD is a low-cost option for less frequently accessed workloads.

8.What are IOPS?
IOPS stands for Input/Output Operations Per Second.
It represents how many read and write operations a storage volume can perform per second.
Simple ga gurthupettuko:
IOPS ante storage second ki enni read/write operations cheyyagalado.
Eng: IOPS represents the number of input and output operations a volume can perform per second.

9.What is EBS Throughput?
EBS Throughput is the amount of data that can be transferred to or from an EBS volume per unit of time.
It is usually measured in MiB/s and is important for data-intensive workloads.
Simple ga gurthupettuko:
Throughput ante EBS volume second ki entha data transfer cheyyagalado.
Eng: EBS Throughput represents the amount of data transferred per second.

10.What is an EBS Snapshot?
An EBS Snapshot is a point-in-time backup of an EBS volume.
It can be used to create new EBS volumes or restore data from a previous state.
Simple ga gurthupettuko:
EBS Snapshot ante EBS Volume yokka point-in-time backup.
Eng: An EBS Snapshot is a point-in-time backup of an EBS volume.

11.How do you create an EBS Volume from a Snapshot?
We can select an existing EBS Snapshot and create a new EBS Volume from it.
The new volume can then be attached to an EC2 instance in the appropriate Availability Zone.
Simple ga gurthupettuko:
Snapshot nundi new EBS Volume create chesi EC2 ki attach cheyyachu.
Eng: We can create a new EBS Volume from a Snapshot and attach it to EC2.

12.What is EBS Encryption?
EBS Encryption protects data stored on EBS volumes and their snapshots.
It uses AWS KMS keys to encrypt data at rest and helps protect sensitive information.
Simple ga gurthupettuko:
EBS Encryption ante EBS lo data ni secure ga encrypt cheyyadam.
Eng: EBS Encryption protects data stored on EBS volumes and snapshots.

13.Why do we use EBS Encryption?
We use EBS Encryption to protect sensitive data stored on EBS volumes and snapshots.
It helps meet security and compliance requirements for data protection.
Simple ga gurthupettuko:
Sensitive data ni secure cheyyadaniki EBS Encryption use chestham.
Eng: EBS Encryption protects sensitive data stored in EBS.

14.Can we increase EBS Volume size?
Yes. We can increase the size of many EBS volumes without detaching them from the EC2 instance.
After modifying the volume, the operating system may also need to be extended to use the additional space.
Simple ga gurthupettuko:
EBS Volume size increase cheyyachu, tarvata OS lo partition or file system extend cheyyali.
Eng: EBS volumes can be expanded and the operating system can be extended to use the extra space.

15.Can we decrease an EBS Volume size directly?
No. EBS does not directly support shrinking a volume.
To reduce the size, we generally create a smaller volume and migrate or restore the required data to it.
Simple ga gurthupettuko:
EBS volume ni direct ga shrink cheyyalem, smaller volume create chesi data migrate chestham.
Eng: EBS volumes cannot be directly shrunk; data must be migrated to a smaller volume.

16.What happens to EBS data when an EC2 instance is stopped?
EBS volumes normally remain available when an EC2 instance is stopped.
When the instance starts again, the attached EBS volume can be used with the preserved data.
Simple ga gurthupettuko:
EC2 stop chesina EBS volume and data normally preserve avtayi.
Eng: EBS volumes normally persist when an EC2 instance is stopped.

17.What happens to EBS when an EC2 instance is terminated?
The behavior depends on the DeleteOnTermination setting of the EBS volume.
If enabled, the volume is deleted with the instance; if disabled, the volume remains available.
Simple ga gurthupettuko:
EC2 terminate ayinappudu DeleteOnTermination setting batti EBS delete or preserve avtundi.
Eng: EBS deletion on instance termination depends on the DeleteOnTermination setting.

18.What is the difference between EBS and EFS?
EBS provides block storage that is normally attached to an EC2 instance.
EFS provides shared file storage that can be accessed by multiple compute resources.
Simple ga gurthupettuko:
EBS block storage, EFS shared file storage.
Eng: EBS provides block storage, while EFS provides shared file storage.

19.How would you troubleshoot an EBS performance issue?
I would check the volume type, IOPS, throughput, volume size, and CloudWatch metrics.
I would also check whether the workload is exceeding the configured performance limits.
Simple ga gurthupettuko:
EBS performance issue vaste Volume Type → IOPS → Throughput → Size → CloudWatch check chestham.
Eng: I would check the volume type, IOPS, throughput, size, and CloudWatch metrics.

20.What is the most important thing to remember about EBS?
EBS is persistent block storage mainly used with EC2 instances.
The important concepts are Volume Types, IOPS, Throughput, Snapshots, Encryption, and Volume Expansion.
Simple ga gurthupettuko:
EBS ante EC2 ki persistent block storage provide chesi data ni store cheyyadaniki use chese AWS service.
Eng: EBS is persistent block storage used to store data for EC2 instances.